



CHAROTAR UNIVERSITY OF SCIENCE & TECHNOLOGY

Seventh Semester of B. Tech. (EC) Examination

May 2013

EC402 RF & Microwave Engineering

Date: 07.05.2013, Tuesday

Time: 10:00 a.m. To 01:00 p.m.

Maximum Marks: 70

Instructions:

1. The question paper comprises of two sections.
2. Section I and II must be attempted in separate answer sheets.
3. Make suitable assumptions and draw neat figures wherever required.
4. Use of scientific calculator is allowed.
5. Use Graph Papers.

SECTION – I

- Q - 1 Explain advantages and general applications of microwave. [07]
- Q - 2 (a) Explain directional couplers [08]
- (b) Explain Faradays rotation principle. [06]

OR

- Q - 2 (a) Explain circulators. [08]
- (b) Discuss about standing wave and standing wave ratio. [06]
- Q - 3 (a) A transmission line has following parameters. [06]

$$R = 2 \frac{\Omega}{m}, G = 0.5 \text{ mmho/m}, f = 1 \text{ GHz}, L = 8 \text{ nH/m}, c = 0.23 \text{ pF}$$

Calculate: (a) the characteristic impedance (b) the propagation constant.

- (b) A lossless transmission line having characteristic impedance 50 ohm, is terminated by load impedance $z_L = 100 - 150j$ ohm. Using smith chart find [08]
- 1) line impedance at 0.5λ
 - 2) SWR and draw SWR circle
 - 3) Reflection coefficient
 - 4) Return loss in db

OR

- Q - 3 (a) Explain reflection coefficient and transmission coefficient of transmission line theory. [06]
- (b) Explain impedance of transmission lines. [08]

SECTION – II

- Q - 4 Discuss power transmission and power loss in rectangular waveguide with necessary equations. [07]
- Q - 5 (a) Explain magic tee and derive its s parameter matrix. [08]
- (b) Explain TE modes in rectangular waveguides. [06]

OR

- Q – 5 (a) Explain E- plane tee and derive its s parameter matrix. [08]
(b) Explain TE modes in circular waveguides. [06]
- Q – 6 (a) Explain principle and applications of magnetron. [04]
(b) Explain operation principle of TRAPATT diode. [05]
(c) Explain microwave strip lines. [05]

OR

- Q – 6 (a) Explain two cavity klystron amplifier with velocity modulation. [04]
(b) Write short note on parametric amplifier. [05]
(c) Explain strip lines. [05]
